RAGHUL S NAIR

Data Scientist | Analytics & ML Projects

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Portfolio: https://raghulsn.github.io/Portfolio

SUMMARY

Mathematics graduate with a quantitative foundation in statistical modeling and optimization, applying these principles to develop end-to-end machine learning solutions in credit risk and customer analytics. Proficient in Python and SQL, with hands-on experience in model validation, risk segmentation, and decision optimization frameworks.

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, Pandas, NumPy
- **Machine Learning:** Logistic Regression, Tree-Based Models, Clustering, Feature Engineering, Model Validation, Hyperparameter Tuning, Threshold Optimization
- **Analytics & Visualization:** Power BI, Matplotlib, Seaborn, Plotly

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PROJECTS

- **Customer Churn Reduction & Retention Optimization Engine** View
  - Built and evaluated ML models (Logistic Regression, Random Forest, XGBoost) on 7,000+ customer records, achieving **ROC-AUC of 0.84** and improving recall on high-risk customers by **25% vs baseline**.
  - Implemented cost-sensitive learning and lift analysis, delivering **3x lift in the top 20% risk segment** and improving targeting precision for retention campaigns.
  - Designed a budget-aware retention optimization framework that increased projected retained revenue by 20% compared to probability-only targeting
- **Probability of Default Modeling & Credit Policy Optimization** View
  - Developed a **Probability of Default (PD)** model using logistic regression on **1.8M+** consumer loans, achieving **AUC 0.71, KS 0.31, Gini 0.42** under time-based validation.
  - Performed **decile lift and risk band analysis**, demonstrating strong monotonic separation (5% vs 46% default between lowest and highest deciles).
  - Optimized credit approval threshold via **expected loss simulation**, identifying a policy cutoff (~0.20 PD) balancing ~65% approval rate with controlled portfolio default (~14%).
- **Sales Performance Analytics Dashboard | Power BI** View
  - Designed and implemented a star schema based Sales Performance Dashboard analyzing revenue, profitability, and YoY
  - Developed KPI-driven growth across ~10K transactions using advanced DAX measures (CALCULATE, SAMEPERIODLASTYEAR, RANKX, DIVIDE).insights including Profit Margin %, Contribution Analysis, Running Total, and regional revenue concentration to diagnose performance drivers.
  - Enhanced time intelligence accuracy by creating and marking a dedicated Date dimension table for reliable period-based comparisons.

  Additional Projects on GitHub | Kaggle

PROFESSIONAL TRAINING

- **NSDC** – Professional Certificate in Data Science & AI *(Avodha)*
- **IBM** – Professional Certificate in Data Science *(Coursera)*

  Additional Certificates on LinkedIn | Sololearn

EDUCATION

- **Bachelor of Science in Mathematics,** *University of Kerala* (2017 – 2020)